The Social Radars

S4: Surbhi Sarna, Founder of nVision Medical

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Jessica: I'm Jessica Livingston, and Carolynn Levy and I are the Social Radars. In this podcast, we talk to some of the most successful founders in Silicon Valley about how they did it. Carolynn and I have been working together to help thousands of startups at Y Combinator for almost 20 years. Come be a fly on the wall as we talk to founders and learn their true stories.

Carolynn, I'm really excited. Today we have on the show Surbhi Sarna, who is a Y Combinator partner, and before that, she was the founder and CEO of nVision Medical, which was the first and only device... It was the first and only device to test for ovarian cancer, and I can't wait to talk about this with you, Surbhi. Welcome.

Carolynn: Welcome.

Surbhi: Thanks for having me. I'm so excited to be here.

Jessica: And also, you published a book last year called Without a Doubt, which I'm very excited to talk about. And the book chronicles your journey through the company. I want to go back to the beginning when you were in high school and suffered from complex ovarian cysts. Tell us that story because that sort of kicked things off for you, didn't it?

Surbhi: It did. It did. I would say that even before I started getting these extremely painful cysts, high school was already an interesting time for me as it is for many people. I was already a little bit science-leaning, a little bit nerdy, and then one day I was writing a paper on Emerson in my bedroom, and mid-sentence I felt this really sharp pain that kept getting sharper. Sometimes something passes, but it keeps getting more and more acute. And my mom was in the kitchen at the time, and I remember getting there, just seeing the silhouette of her back against the window and then passing out.

Jessica: Oh my gosh.

Surbhi: Yeah. The next thing I remember is waking up in the car for a second actually and thinking, "How did my mom get me here? How did she get me in the..." Because I wasn't a mom at the time, but now I know.

Carolynn: Superhuman mom strength.

Surbhi: Yeah, yeah. Your kids are passing out on the floor, you're going to get them in the car.

Carolynn: Exactly. Yep.

Jessica: So at the hospital though, they couldn't properly diagnose if you had cancer or not. Tell us about that because it's really scary-sounding.

Surbhi: Yeah, it was... The ER visit that day was excruciating because I was in so much pain. I realized that I could stop myself from passing out from the pain if I kind of hunched over a bit, if I never stood up straight, which is kind of an odd thing, and at first they thought that I had appendicitis. It turns out imaging for the appendix isn't that great either, which is so interesting because it's a common procedure. And they prepped me for surgery, and then only right before the surgery the surgeon realized, this isn't appendicitis, this is something else, but we don't quite know what it is yet.

Carolynn: They were actually going to open you up and remove your unburst appendix? They were just going to open you up and-

Surbhi: Yeah.

Carolynn: Wow. Wow.

Surbhi: Yeah, the appendix, it turns out that there's all this medical error around appendicitis I had no idea about until recently when I had a girlfriend who they missed her appendicitis and then she ended up in the hospital for more than a month because the infection spread everywhere. so even things that are quite routine, there's some-

Carolynn: Actually, maybe getting your appendix out proactively based on that story doesn't sound like such a bad idea.

Surbhi: I know, I know. Yeah. I was scared that day. I was 13 years old, but luckily, the surgeon right before... He came in, and I remember he was masked up and everything, kind of doing one last check, and he had me stand up on my toes and come down on my heels as hard as possible, and supposedly when you have appendicitis, that should kind of make you feel excruciating amounts of pain, but that didn't trigger it for me that time. They kept me for a couple of hours for observation and then pulled me back, and then only the next couple of days realized that I had very large complex cysts on my ovaries.

Jessica: Through a scan?

Surbhi: Yeah, through a transvaginal ultrasound.

Carolynn: Yeah, yeah. I was going to say. Okay.

Surbhi: They don't readily do those on very young women, so I think that was part of it. It took a couple of days to say, "Okay, to know this, we're going to have to do this other procedure." But then the issue with the ultrasound is you can see that there's something on the ovary, you could see that there's a mass, and sometimes if that mass is completely fluid-filled, then you know that it's nothing to be worried about. It's just a cyst. Go home, let it pass. It's fine. But the reason they call the mass I had complex is that it's partly solid, partly liquid. So when there's that sort of thing, you can't really tell.

You don't want to rush to operate because the woman's so young. So then the next thing they do is a blood test, and unfortunately, this blood test till today is extremely inaccurate. So Kaiser, for example, won't even do it in women who are premenopausal.

So these lack of diagnostic options, this is why in the US, according to the Cleveland Clinic, there's 600,000 operations to remove the ovaries of women in the US each year, but there's only 14,000 cases of ovarian cancer and still three-fourths of those are caught at late stage. So we're jumping to surgery and then often there's nothing there.

Carolynn: Wow.

Jessica: Because that's the only sort of safe option?

Surbhi: You can't just biopsy the mass when it's sitting on the ovary because the ovary sits in the middle of the rest of the cavity. So imagine if you go in with a needle and take something that's cancerous and you pull it out, you're basically spilling it into the rest of the cavity. So they really don't want a biopsy if they can prevent it, and often when they try a biopsy, they end up removing the rest of the ovary, and doing that at 13, 14, truly almost at any age is not great. It's not great for women's health just to pull out the ovaries.

Carolynn: Did you have cysts on both or just one?

Surbhi: On one.

Carolynn: Okay. So what can happen, and maybe the case that you take out one, but you can still keep the other one and still be fertile?

Surbhi: You can. You can still be fertile. The ovaries are also sort of part of what controls our hormones. So in addition to the fertility part of it, women that have, especially both ovaries removed, suffer from a higher risk of osteoporosis, or heart disease. You want to proceed with caution.

Jessica: So back then at 13 though, you didn't know all this stuff. You were basically told, "We see a mass, it could be cancer," it might not be, and you and your family had to make the scary decision of what to do, which was I think you said, let's not operate and wait.

Surbhi: Yeah, that's right, and this was all happening back in 2004, and Google really wasn't a thing yet. When I was speaking at start-up schools to crowds of kids, I'm like, "I did research on Yahoo and actually went to the library." I had really spectacular teachers, and so they had a lot of patience with me in terms of helping me understand the scientific concept behind what I was going through, and I think equally importantly, I would say, and I think this is what led me eventually to writing the book, I had an English teacher named Ms. Middlestead that I was really close with, and she taught me how to channel the emotions I was feeling into writing to keep me productive.

And I started with writing, almost making fun of certain aspects of what I was going through. For a follow-up, transnational ultrasound, they told me I should drink a lot of water and not pee before, and I think I took that to an extreme. And so writing can help you process really difficult emotions, and it can help you see things in a more lighthearted way as well when you're looking back on it, and I think a combination of those things got me through. And of course, the support of my mom got me through that time.

Carolynn: How did they manage your pain?

Surbhi: That's a great question. I can't remember if I was on anything prescription-level, but I do know I was on painkillers around the clock because eventually there's a limit to how much school you can miss, and if you feel like you can go back, at least you have some sense of normalcy.

Jessica: Did it ultimately correct itself or what happened to the complex cyst?

Surbhi: Yeah, eventually after six months... I know this is not a pretty word, but it bursts, and that's how it kind of corrects itself. When it does, it causes torsion in that area, things kind of twist together. So I remember I was in Spanish class when it happened and it was extremely painful, and then all the teachers were alerted to what I was going through clearly, and then I had to remove myself and make it to the nurse's office. And of course, the teacher doesn't want to speak in English because it's Spanish class, so she tries in Spanish to explain to the other kids that I had a mass on my ovary which ruptured, to a group of teenage boys.

Jessica: Oh my word.

Carolynn: Oh, as if you learn the word for ovary in Spanish. That's just so absurd.

Surbhi: So I had a boy run up to me later and say, "Ew, I heard your ovary burst." I was like, "It's not my ovary. Okay, never mind."

Jessica: Oh, that's so hard, especially as a 13-year-old when you're so sensitive about everything. But once it burst, then your body naturally sorted it out and you were okay?

Surbhi: So after that, I would have sort of a cyst every year and every time you have a cyst, there's some actual increase in the chance that it was cancer, but it actually made us less nervous each time because we knew what we were dealing with, we knew how to monitor it, we knew how to watch it. But going through something like that, even though I was at a progressive, very supportive school, I just realized that historically there'd been this difference between the way that women's health was prioritized, and it kind of started this bug in me that one day I wanted to start a company in the space.

Carolynn: Okay, but let's stop there. I think this is really interesting. You were thinking, "I want to start a company in this space," but why aren't you thinking, "I want to go to a prestigious university and be a researcher?" How did you know you wanted it to be a business versus just like I want to be in a lab and discover the thing that helps women, this particular issue-

Surbhi: Yeah. My dad was an engineer by training. He's a software engineer, and he was one of the first few hundred people at Netscape just as an engineer working there, but oh my gosh, he loved it. He would talk about it all the time, the startup thing. He would wear the swag, the jacket. I remember the day that Netscape IPO'd. So we were in the middle of this tech innovation burst.

Jessica: Yeah, that's ground zero back then, Netscape. Yeah.

Surbhi: Yeah, and Netscape and Microsoft were warring with each other, and Netscape left a giant Mozilla thing on Microsoft's ground, and Microsoft put an M on top of it, and I saw how energized my dad was being part of this after working at very large companies before that, and I just love the concept of finding and building new things. And even before I got sick, when I was a kid... I still have the notebook where I would, I know it sounds so nerdy even today, but change the habitat of different insects and see how they would react to stuff. My parents got a treadmill and at the age of six or seven I was trying to assemble it myself and put it together. So I think it's a really good question, why didn't I just say immediately I was going to go into research? And I always kept that part of that option open, but that excitement, being a Bay Area kid, growing up in that time, watching my dad be so happy being even a smaller part in creating something like that, I think I just got infected with the bug a bit.

Carolynn: Was your mom entrepreneurial too?

Surbhi: So my mom has a PhD in Hindi literature. So she's not traditionally entrepreneurial, but she always wanted to teach Hindi. That was her goal. And then she had an interview lined up at UC Berkeley, and then she found out she was pregnant with me and decided to hang back from it. I'm super proud of her today because she's actually teaching Hindi at Stanford.

Jessica: Oh, wow. That's amazing.

Carolynn: That's awesome. It was delayed, but she eventually reached her goal.

Surbhi: And it's amazing now to see her photos of her on the Stanford campus, and in fact, I didn't even know I funded one of her students, and so-

Jessica: Really?

Surbhi: ... they came up to me. Yeah, they're like, "Are you Anupama's daughter?" And it was really cute.

Jessica: That's wonderful.

Carolynn: Love that.

Jessica: So then you went to Berkeley, then you undergrad, and then after you got more of a traditional job, I think it was at Abbott Vascular and another med tech type of startup.

Tell us about the moment when you decided to start a company to make a device to test for cancer.

Surbhi: I was working this other job, I was getting a lot out of it, and I always felt like I needed a certain number of years of experience before doing a company in healthcare. So I was 24 at the time when my grandmother's breast cancer came back, and between Berkeley and working my first job, I had spent the summer in India helping to take care of her. That's a very different experience studying cancer academically, even having a scare and taking care of someone firsthand who's suffering from the disease, right?

Jessica: Yeah.

Surbhi: But when she passed away when I was 24, I had just gotten engaged and I thought she was coming to my wedding, and I think at that point I just realized if this is something I really wanted to work on, I had understood the disease so well at that point I had read every paper I could get my hands on that I just needed to dedicate myself to it fully if I was going to discover something. So I think that... And because I wasn't doing that, I was engaged to my best friend and I was doing well at my job, but I just wasn't satisfied. There was something in me that wasn't satisfied and the joy that I got from then focusing on this problem and knowing that I was just going to spend every ounce of myself trying to find a solution, the joy was instantaneous. Even though it was so hard and so uncertain, that part of me, whatever was missing, was solved instantaneously. I don't even know how to describe it accurately.

Carolynn: Now looking back, you're like, "This is what I was meant to do." No reservations?

Jessica: So what was the first thing that you did though? Was it like, "I'm going to sketch out what it'll look like."? I mean, how did you even know because your device goes in through the fallopian tube and tests those cells. Did you even know that that was what it was going to be?

Surbhi: So there was this paper published by a lab out of Johns Hopkins, the Vogelstein Lab, that said ovarian cancer is actually a misnomer. We should call it fallopian tube cancer because the majority of ovarian cancers actually begin in the fallopian tube. That's where the most cellular turnover is happening. It was like the seminal moment in understanding this disease, and as I had studied every other cancer to figure out how have we as a society really made an impact on cervical cancer, you have the pap smear; on colon cancer, you have the colonoscopy; mammogram for breast cancer. Really we have to get to the source of origin. That's what's really made the difference wherever we've made the biggest difference in whatever cancer, and that's not always possible. You can't easily access the pancreas, for example. But I thought, "What if you could easily access the fallopian tube?" So I started reading all of the FDA filings, all of the papers ever written, truly ever written, on anyone that tried to access the fallopian tube or ovary for any reason. That's where I started.

Carolynn: Without having to open you up, yeah.

Surbhi: No incisions. What have people done in the past? I don't want to reinvent the wheel. What have people done in the past? What can I learn from it? And I learned that there had been multiple attempts to get to the ovary for other reasons. One of them being imaging of the fallopian tube for infertility workup. If there's a blockage in the fallopian tube, if you can get a catheter up there, you'd actually be able to visualize the blockage and treat it, and that's one of the leading causes of infertility. And so that's where I started reading everything I could and then actually sketching out the idea. And then eventually I knew that I needed to talk to physicians about it to make sure that they thought it was a good idea as well.

Carolynn: Are you still working at this point? So you're doing all this at night, in your spare time, but still have a job?

Surbhi: I still have my job, yeah.

Jessica: But you start talking to a gynecologist that you knew?

Surbhi: Well, yeah. I mean, this is interesting, right? I so wish I had YC on the journey. It was right before YC was doing healthcare stuff, but I just felt intuitively I must talk to as many users, as many physicians as I possibly can, and it is not like you have a huge Rolodex when you're young, wanting to start a company like, "I can call 10 physicians and see if what they think of this."

Carolynn: You can call your own gynecologist.

Jessica: Which is what she did.

Surbhi: I sure did. I called and first I was like, "Hey, I have a new catheter concept I've left to talk to Dr. Cook about," and of course they think you're someone soliciting them for something, and I got hung up on by the front desk. So I waited a week, feigned a bellyache, got seen immediately, and I kid you not, that's like the definition of putting yourself out there to talk to users. She had my feet up in stirrups and then when she was peeking up from the curtain, I was like, "I have this catheter concept I want to tell you about. I want to be a founder. Listen, listen."

She was like, "Not right now. Not right now please. But if you stop talking to me about this right now, I will meet you for a coffee at a nearby coffee shop," and that's what she did. And then she liked the concept and she introduced me to the head of women's health at Stanford at the time, to US Davis. Yeah, and then all of a sudden I had this network of going from nothing to this network of physicians who thought, "Hey, if this idea was in the world, it would make a big difference to patients."

Carolynn: At this point, how conceptual is it? Was it a working prototype and that you knew it would actually work, or was it more like this is a concept and I just want to hear what gynecologists think of it?

Surbhi: Just conceptually because it was a brand new concept. Hey, what if a woman during her annual exam, there was a procedure you could do, especially if she was high risk for developing ovarian cancer, there was a procedure you could do that would grab these cells without making an incision? How would you react to that? Is that something you would want to do? Even just making sure that folks were like, "I would never do that for these reasons." And then the negative feedback I got was as valuable as the positive feedback. I mean, that's where I learned the most about as I went into the design of the product, what were some of the key things it would have to embody in order to be picked up by a gynecologist.

Jessica: Which is what? Tell us.

Surbhi: The simplicity of the procedure was incredibly important because there's another endoscope or visualization. So an endoscope is basically a camera that used for medical purposes, and there was another camera that they're already using in office to take a look at the uterus, and so when they're looking at the uterus, they could then see the openings of the two sides of the fallopian tubes, but at that point, they're not looking at the patient, they're looking off at the screen and they're using their hands like this in front of them, but they're not looking at their hands. So if you don't make something that's intuitive to use, they have to take their eye off the ball to look down at the device, they're going to be prone to making accidents.

So those were the type of things I learned early on coming from a very patient-centric viewpoint, being a patient myself, that being the motivation of why I wanted to start the company. So these were the things that I learned by just talking to them and iterating with them every way. Even when we had a handle design. You can imagine that for a long time when most innovators for physicians were men, and when most physicians were men, the handle of medical devices fit the hand of a man. Not a woman, not a female physician. And so that was something where even the way that our handle was designed, there were two different ways to grip it depending on your hand size.

Jessica: Oh, wow. Okay.

Carolynn: Yeah.

Surbhi: So there was just a lot of attention to detail like that that came from being in constant touch with our physicians as we were developing this thing.

Carolynn: Is there a way to test for uterine cancer also? We know the pap smear is cervical, but how do you get cells for uterine cancer? Could you have done something for uterine cancer as creative as you did for ovarian at the same time, or is it like no, you have to-

Surbhi: No, I did try to do both. That's a great question actually, and what we ended up designing... After years and years of trying and failing, what we ended up with was a catheter which was self-navigating. So if you think about other devices that go into the body, like stent placement, the physician has to kind of watch the screen the whole time and actually navigate the catheter into the right spot into the heart, and then deploy the stent, tight? The fallopian tube is so tortuous, there's so many twists and turns, and it's only a millimeter in inner diameter. So we're talking like tiny, tiny, tiny. So anything else that anyone had invented at that point would go into the fallopian tube and puncture it, because it was too stiff because there was a small air in navigation. The other thing about the fallopian tube, unlike your coronary vessels which stay open naturally, the fallopian tube lays flat. It's a potential space. It needs to be pressurized to open.

So it was crazy complicated to try to invent something. What we created was a linear averting balloon catheter, which if you imagine pulling a sock out of the dryer and then trying to straighten the sock. Okay, the way the balloon sits inside the catheter is like this, and then it slowly unfolds. Then we put a unique texture on the outside of the balloon. So you could think about how you get a cheek swab, there has to be some friction, but if you put too much friction in, you're going to damage the tube, and I didn't want to do that either.

So we were the very first people to create a textured balloon to grab cells from any part of the body. So we created a textured balloon that would unfold into the fallopian tube, grab cells, but then you would pull it back into a sheath that was sitting in the uterus. So my idea was always, as we're looking at unique diagnostic assays, we can look at what's on the balloon to prevent contamination, because you don't want it to touch any other part of the body. You pull it back into this outer layer, the sheath, but the sheath is making contact with the uterus.

Jessica: Yeah. So you could use something with the sheath to test that?

Surbhi: Yeah, exactly. So in my brain, where I wanted to take the product long-term was a full panel for women's health, cervix, uterus, fallopian tube.

Jessica: Wow.

Carolynn: How far did yourself self-navigating catheter have to go in any given fallopian tube to get the cells? You know what I mean? Could you just go into the beginning of it or did you have to go really far into it?

Surbhi: We did clinical studies to answer this question. So it's a great question. So at the very end of the fallopian tube, you have something called fimbria, which are like finger-like extensions that sit on top of the ovary. And our bodies are so incredible, outside of a certain time of the month, the ovary actually physically pulls away from the fallopian tube so that if you do have an infection, it doesn't reach the ovary.

Jessica: Oh, wow.

Surbhi: But when it's ready to... Yeah, it's very dynamic. But when it's ready to capture whatever it needs to, it actually comes back. But the finger-like extensions is actually where the cancer is most likely to start. So getting as far in as you could was important. The basic literature on how long the fallopian tube is on average was missing.

Carolynn: I was going to ask you that, because we all talk about how long the intestines, and how long the colon, but how long is a fallopian tube?

Jessica: How do they not know that? How has no one researched that? It's like reproductive health.

Carolynn: Because women's health is not as important.

Surbhi: Yeah. Just not a priority for a long time.

Jessica: Oh my God.

Carolynn: How do you build a catheter when you have no idea what the length that needs to go is? That's crazy.

Jessica: I'm feeling overwhelmed on your behalf actually thinking this, and I have a lot that I want to get through actually, but this is so interesting. You've mentioned the FDA. Is it too quick if we jump to what was the process of getting FDA clearance?

Surbhi: A big part, maybe the most important part, I'm not sure, of building a company in the early days is who you surround yourself with, and there was one woman named Cindy D'Amicus who had the best relationship with the FDA in terms of women's health. In fact, she had sat on a panel on behalf of the FDA in the past, that sort of thing. But she was so busy that she was one of those folks that if you wanted her time, you would have to pitch her, and it's still like that today. Now I'm pitching her on behalf of YC companies. I'm like, "Come on, take them on and let me tell you why this is so important." And she met with me a couple of times. I explained the concept, and she had a dear friend pass away with a related a woman's cancer that always wanted to do something in this area, and I think the combination of that and meeting with me a couple of times, hearing about the concept, she took me on as my partner to think through the FDA process.

Jessica: Wow. That's awesome.

Surbhi: I was so lucky. I got so lucky with the people I got to hire at nVision. I mean, my first technical hire was my previous boss. He was the VP of R&D at the other company I just worked for, and I remember telling him when I was leaving, "I want to start a company," and he said, "Okay, just remember me when you start Cervico," and this is part of why it's good to give everything you have to any job that you're in or anything that you do, because here, I think I'm 20, 25 years younger than him, and he already had several exits under his belt. Now, all of a sudden, this entry-level engineer is coming to you and saying, "I started a company." But he just remembered, you worked so hard, you were so creative. There was something about our interaction where he said, "Okay, fine. I'll give this a shot."

Jessica: Actually, I want to put a pin in the FDA thing because I now am reminded that I wanted to ask you about being a solo founder as you're talking about your team and how the importance of bringing on amazing people. There's a misperception out there that Y

Combinator doesn't fund solo founders, and that is not true. We have always funded solo founders, male and female. You are a solo founder. Tell us about that experience.

Surbhi: So it took me a year and a half to raise my first $250,000. This was in 2009, the market wasn't great, and now there's pretty words like FemTech to describe women's health. Women's health wasn't a category. I would go pitch it and people would say, "Oh, sorry, I don't do bikini medicine."

Jessica: Are you kidding? Ew.

Surbhi: No, no.

Carolynn: I never heard that either.

Jessica: Bikini medicine? That's super lame.

Surbhi: Yeah. It took me a second to get it. Oh, anything that a bikini touches, I get it.

Jessica: Yeah.

Surbhi: Okay. So first of all, a lot of the other engineers I worked with early in my career were men, and women's health was not a category at that point. It's just we were living in a very different world now. It wasn't like this category that had a lot of focus, and if there was ever a women's health company, it did poorly. So all of the comps were bad. And this is why I'm always really careful about too much pattern-matching. It's like, well, how are you going to spot the next thing if you're only funding the thing that has already worked? It's dangerous for a variety of reasons, but in this case, there was no big women's health company that was created anytime in the previous 10 years.

And so I think it was... I found a couple of guys who were like, "Okay, I'll give this a shot. I'll give you some hours in the evening," but for them to get super excited about this product, I think it was just more difficult. I did approach some women engineers as well, and I think that maybe there's this dynamic, and I'm just guessing at this point, it's really hard to get a comfortable job as a woman engineer, especially back then. So this whole idea of like, okay, yeah, you were the only woman in all your engineering classes and now just leave everything, come do this thing that no one's funding yet, but it's going to be great. It was really hard to find that.

Carolynn: Oh, totally. Because they're sitting there thinking, "I want to break the glass ceiling at the place where I got this job, and so for me to leave and go join you means I'm not going to be able to accomplish that goal." So I can totally see why it was hard to get women on board.

Surbhi: Yeah, and it's scary. I remember being at the larger company and walking down the hallway and being surrounded by glass with the offices on either side, and it's like men, men, men. You know what I mean? One woman. So it already seems like things are set up to be really difficult. So yeah, at least let me stay here and break this glass ceiling. Where's the example of success, the female founder? It's just layers of complexity.

Carolynn: Yeah, layers of barriers.

Surbhi: But I was going to do this no matter what. That was the thing. If it took me a year and a half to fundraise, if I didn't have the right partner, I was going to make it work. And now when I fund both... I did two solo founders last batch, and I get to see firsthand the benefit of having someone there to help you carry the load. Emotionally, if you're out fundraising things don't stop operationally, and I see my solo founders now, how burnt out they get because how do you fundraise around the clock and also keep customers happy if you're already out in the market? Eventually, the first couple of execs that I hired full-time, I really trusted them, and they stayed with me through the early days of the company all the way through two years into the acquisition. So they were a huge help. But the solo founder thing is tough. It's tough.

Jessica: And now that you are funding companies, you can see the difference between the solo founders and ones with co-founders?

Surbhi: Yeah, the solo founders get beat up.

Jessica: Sorry, I shouldn't laugh. It's rough though.

Surbhi: Yeah.

Carolynn: There's just so much work.

Jessica: I know. There's so much work to be done.

Surbhi: There's so much work in the early days. There's a lot of founder work. Founders have to be talking to customers, founders have to be talking to investors. So yeah, I think in the early days especially, it's really rough.

Jessica: You talk a lot about fundraising in your book. I want to talk about this. The theme is rejection. I will say you join a list of impressive people who have been on this podcast that had tons of investor rejection that I just... Carolynn, we kind of can't believe it when they tell us, but wow, when you said it took a year and a half to get 125k. Tell us your... Oh, 250.

Surbhi: 250, yes. 250.

Jessica: Still, a year and a half.

Surbhi: I mean.

Jessica: Tell us about your fundraising experience.

Surbhi: I mean, I get it. I was young. I didn't have much experience. I didn't have a higher degree. To the point earlier, I always kept the door open. Maybe I'll go back and get a PhD, maybe I'll go get MD, but for most people doing not just health tech, even in health tech, there's some expectation that there's a longer time spent in healthcare or a higher degree, that sort of thing, but this is kind of hardcore hardware, medical device, traditional, traditional healthcare. So to be a young woman without a higher degree, without much experience, man, I got the door slammed in my face multiple times. But if they slam the door politely, that's fine. I just remember one prestigious fund, I had made it all the way to their partner meeting. So there was actually one guy there who took me really seriously, and he said-

Jessica: Wow, partner meetings. That's impressive.

Surbhi: Yeah, and he's like, "Wow. Look at the talent you've attracted to the table. Your IP strategy's sound." He took a real close look at it. He talked to physicians, "Okay, you're going after a real problem." And he is extremely successful himself, so you think he would also demand a certain level of respect from his partnership in addition to just being courteous to me, but that's not what happened. They started talking to each other like I wasn't in the room. Yeah, I'll never forget it.

Jessica: Were they talking about your idea or were they just chit-chatting?

Surbhi: They were chit-chatting and they were on a screen, and I don't know, they just started showing each other stuff on their phones, talking about another deal, and then they looked over and they're like, "Oh, we should probably mute this," and they mute it, and I was in the middle of pitching my heart out.

Jessica: Oh my God. I am so appalled at this story.

Surbhi: Yeah.

Carolynn: Yeah, me too. That's just bad behavior all around.

Surbhi: Yeah, it's bad. It's bad behavior. And part of these experiences I think is why partners at YC are so protective of our founders, because we remember what it was like to go out there and get beaten up and what that did to us emotionally and what it took to kind of rebound into a place of confidence for that next pitch, which I closed the next pitch right after that. But it was rough. New category, young founder. I used to say that I got rejected 50 times, but I wrote the book but part of it was rereading, I don't even know how many emails, and I think I just stopped counting after 50.

Jessica: Oh my gosh, that's so... Oh, I get so mad when I hear that because like you said, if they rejected you politely, you'd be okay with that because you did say at one point, I was reading, you said, "You don't have to make every investor love you. You just have to make a couple," and that's fine, and not every investor's going to want to invest in every company, we all get that, but to be rude and to be having separate conversations when a founder is there by themselves pitching their heart out, just... Oh, it makes me so mad. But you finally did convince two investors, right? And you scraped together 250k?

Surbhi: Yeah, this is also kind of a funny story. Okay, so there was this woman named Anula Jayasuriya, who is from the Bay Area, loved the concept, and she was making all of the introductions to the various potential investors. She introduced me to this other woman named Darshna who was based out of Boston, and Darshna, she talked to me for a full six months before inviting me to her partner meeting, and it was basically the last investor on my list, on my friends of friends list, on Anula's list. We had exhausted all of the other channels.

And I remember going into that partner meeting like, "This needs to be the best pitch I've ever given in my entire life. I have to just knock this out of the park," and I felt like it did. And then I felt like they were really what I was working on, they were answering all the right questions. Just a feeling you get when a pitch is going well, and Darshna walked me to my car after, and she said, "Have a safe flight back to San Francisco. I'm going to call you tomorrow with some good news," and it was really hard not to count chickens at that point.

Jessica: With some good news? For sure.

Surbhi: Yeah, and you know what? I think she was really optimistic about how it went too. So I went home and I set my alarm early so that I'd be awake East Coast time for this phone call, and she calls and she just says, "Listen, I couldn't get my partnership over the line. You're just too risky of an investment," and oh gosh, it was so rough. It's so funny, these really rough moments, how you remember all of the details. I got off the phone with her, I jumped back into bed, I threw the covers over my face, and I thought, "You know what? I gave it my all." But even me, I have to know when the end's the end. I did everything I could. There's nothing left, Surbhi. Just go get a normal job. And I think there's some value in letting us sit in our emotions sometimes, because once I got past that initial feeling of rejection, I realized that she really wanted to say, yes, actually. She really wanted to do this deal.

I was like, "So what can I give her to push her to empower my champion and just get them over this line?" And so I called Darshna back and I said, "I know I'm a risky investment. I know that, but if you're willing to take a chance on me, I will take a chance as well. I will not give myself a salary for two years, and I will move back home with my parents," which is probably something I should have discussed with my parents first, but...

Carolynn: You were married at this point?

Surbhi: I was engaged at this point. I should have probably talked-

Jessica: I think he might have an opinion about that.

Surbhi: Yes.

Jessica: Dear fiance.

Surbhi: Just a standard founder thing. In that moment, I'm like, "Oh, how do I solve this problem in front of me?" And then luckily everybody was supportive. So I called Darshna back and I tell her this, and at this point, she goes back to her partnership and she says, "Either allow an exception for my contract and let me invest as an individual, or let's do this as a fund," and you know what they said at that point? They said, "Okay, fine. We will give 125,000 if you find somebody else that's willing to give another 125,000."

Jessica: Oh, I hate that bullshit. I hate that stuff.

Carolynn: Classic trick.

Surbhi: Well, I mean, Carolynn got rid of this. See, this is why I love YC so much. With the YC safe, the YC culture, that's out the window now. Yeah. So that's why when I came to YCA, I was like, "Oh, I get why. This is it."

Jessica: They were Boston--based? This is a Boston-based-

Surbhi: They're Boston-based, yeah. Yeah.

Jessica: But you managed to get someone else, right? Didn't you?

Surbhi: Yeah, I managed to get a woman who used to be the president of Edwards Lifescience. Her name's Corinne Nevinny. She's really well respected. So together they put in 250 when nobody else was willing to at all. So I'm still very grateful. And then I met Tim Draper, and Tim Draper was the first man who invested. He matched the 250. So I had 500,000 to build a prototype, and I was off to the races.

Carolynn: I was just going to say, no disrespect, but 250 does not sound like it would've gone far at all in a medical device company, but you got it doubled and 500k fit your plan for how you could get a prototype?

Surbhi: Yeah, I was able to do enough bench testing and then show in animal tissue on the bench. I had all my investors there, check it out, worked with the 500,000, and then the next round after that was four and a half million.

Jessica: Because you had delivered on this prototype.

Surbhi: Yeah.

Carolynn: What was the timeline between the 500k investment and then being able to show them it works and getting another round of investment?

Surbhi: January 2012 was when the first 250 closed, and then that summer was when I got the additional 250, and then April of 2013, so that's a year and a few months later, is when I had the prototype working and raised the four and a half million.

Carolynn: Oh, that's great. That's not that long actually.

Surbhi: No, we did everything superfast. Yeah.

Carolynn: Wow.

Jessica: And did you get FDA clearance before you got acquired in 2018 by Boston Scientific?

Surbhi: I did. So that four and a half million that I raised, we completed two different clinical studies and got FDA clearance with that much money, and that's something I try to bring to YC with the medical devices, but also the biotech that I fund now. There's a lot of bad behavior from that industry that I try to... Because I was so young. So I was coming at this from first principles, and actually if you do that without some of the other learnings from the industry where you need to hedge and you need to try six different prototypes and bring them forward, or five different drug candidates that bring... Let's raise a hundred million dollars when we could be doing this in 10, that sort of thing, if we did it sequentially. My first FDA clearance was in 2015. So yeah, 4.5 was the 2013, and I finished the two clinical studies. 2015 was my first FDA clearance, and then 2016 was my second FDA clearance, and that's when I raised $12 million.

Carolynn: Oh, okay.

Jessica: Wow.

Carolynn: And was that for production? What was the 12 million? What was your plan for that?

Surbhi: It was for early commercialization and to do one more clinical study that would bolster the claims we were able to make to physicians. So the FDA clearance was for collecting a cell sample from the fallopian tube and then looking at the cells for malignant features. And we wanted even more. So we wanted to do one clinical study in patients who had cancer to demonstrate that we could also pull out cancerous cells, and what happened at that point is we did it five times. So it was like a 50 patient study, five times there was cancer in the fallopian tube, all five times we got it. And because these patients were known to have ovarian cancer, as soon as we got the sample, they also had the anatomy removed.

So we were comparing ourselves to the absolute gold standard, which is cross-section examination of tissue. So it was as difficult as it could be, the clinical study design. So we did that five times and I thought, "Okay, great. Let's publish this paper. Let's make some claims and let's push this out to some sites," and my idea was always to take this all the way, because again, being young, starting the company from the place that I did, and plus no one has showed any real interest in anything to do with women's health, I had to claw for everything. So I was very, very surprised when the larger companies, after seeing that five patients' worth of data, started reaching out to me.

Carolynn: So how many did?

Surbhi: Oh my gosh. I think seven. Seven to start with, and it was-

Carolynn: So it was all inbound like, "We saw the paper, and oh my God, you built something and we're super interested."?

Surbhi: Yeah, there's a big medical conference for gynecologists called ACOG, and I just remember being like, "What's going on here?" In May of 2017 where... I thought I was maybe meeting with one or two people and they would bring 10 people from their team, and in medical devices, especially later stage for the next round of funding that I would have had to raise to actually commercialize maybe $40, $50 million. At that point, often the strategists write a big check. And so the reason I was engaging with them was because we had this great data. I was getting ready, maybe in the next year I'd raised this big round of funding, and the acquisition thing just didn't... It felt so early. We're pre-revenue, we have this five pages' worth of data. I had no idea the way things were going to go.

Carolynn: Well. So psychologically... I'm sure there was a lot of technical stuff that happened between expressing interest and ultimately getting acquired, but psychologically, this is your baby, this is your company, you've had all this struggle. How did you get there like, "Okay, I can sell it."? What were all the factors leading up to that?

Surbhi: So right when we got FDA clearance, before I had demonstrated anything in patients with cancer, another company had approached me about an acquisition. So this was before I raised that $12 million round. And I remember taking a walk and sitting behind the AT&T Giants ballpark in San Francisco and looking at the water with a dear friend of mine, and it was life-changing... That acquisition at that time would've been a life-changing amount of money for me because I was supporting my mom, my parents were recently divorced. My husband and I were living in a 700 square foot, one bedroom in Soma, which is right next to-

Jessica: Wow.

Surbhi: Yeah, which is right next to Subway. And to save money, we would go get whatever sandwich was a foot long of the day, because every day they rotate, they have a different sandwich on sale for a foot long for $5, and then we would split that instead of getting-

Carolynn: $5 foot long.

Surbhi: Yeah, $5 foot long.

Jessica: Whatever it was, that's what you'd be eating that day for lunch.

Surbhi: Yeah, exactly. That's it. Yeah, because that way we're spending 2.50 each on lunch. I'd been grinding at that point, and we've been so frugal and so careful, and I was having to make this decision about whether or not to sell, and I just thought... I was talking to my friend, Saneet, and I said, "This next step of showing that we can do this in patients with cancer is so important. I'm not ready to let go yet. I have to take it to this next step," and then of course, Rajiv and I had long discussions about it. And by the way, the company that wanted to buy me, it was a smaller publicly traded company with this amazing female CEO. She was just like, "I'm going to take care of this. You're going to be running a new medical device division," because it was a diagnostic company, "And this will be under your control," and I believed her and I trusted her.

Jessica: That's appealing.

Surbhi: Yeah, it was a big decision, but ultimately I said no.

Jessica: Yikes. That's crazy.

Carolynn: It is crazy.

Surbhi: And then two years later is when the flurry of interest happened. And so I had already had a taste of thinking about it. So we had only raised 17 and a half million total, and a lot of the people that invested in the 17 and a half were smaller funds. So I actually said no even then to the first couple of offers that came in, 100, 150. No, no, no. When it hit 275, it was like the investors were really, really interested, and I had really gotten to know the Boston Scientific team and their culture and just thought, "What are the chances that I'm going to be able to raise this huge round of funding?" Commercializing a medical device is no joke. And that's what I started taking it really seriously, but still the emotional transition over. I stayed with the acquisition even though I was fully accelerated at the time of acquisition.

Carolynn: So you stayed to sort of shepherd the product to where it got before? Did it get all the way to commercialization while you were there?

Surbhi: Yeah, this is a... So I headed up the commercialization effort, and for a medical device, because it's a physical device, what you do is you pre-sell to different hospitals, and a lot of these hospitals, they're on yearly contracting cycles. So what you want to do while they're in the contracting phase is when you want to sell them. So we'd have salespeople on the ground, and then when we got close to someone saying yes or signing on, they would send me in to close the deal. And I've never traveled on a private jet for fun, just for personal stuff, but the very first time I did and just the shift, I'm eating the same sandwich every day to save $3 to being on this jet where they're flying me to five different cities a day to tell... "Go close this $5 million deal. Go close to $10 million."

Jessica: That's a big shift.

Surbhi: So it was a big shift, and it's like all of a sudden I'm presenting big rooms of physicians and I'm closing these deals. So we were definitely all in on commercial mode. And I was just in Boston, and someone asked me, "When was the last time you were in Boston?" And the last time I was in Boston was January of 2020.

Jessica: Oh my God. Right before COVID.

Surbhi: Yes, right before COVID, and we are planning the sales kickoff meeting, like the big launch, because when you launch a device, it's a huge party, like a big deal because you've pre-sold all these contracts, and it's not quite as easy to iterate on the product. And we were planning the launch of it April 2020.

Carolynn: Oh, man. That's brutal.

Surbhi: They're like, "Okay, see you later. Maybe you can come back in March. We can look at the final readout of the clinical study, and then April we'll do our big thing," and then of course, parallel universe, I'm actually flying back to Boston for YC for startup school in 2024 instead of ever being able to go back to launch the product.

Carolynn: To launch the product. That is wild.

Jessica: Oh my gosh, that's crazy.

Carolynn: What a crazy conclusion to that journey.

Jessica: Well, I think people need to read the book without a doubt. And I just need to mention the book though. The cover, the way the typeface is with a doubt is in regular typeface and sort of out is sort of stuck in there as if someone had handwritten it, which suggests despite all of this rejection, Surbhi, you made this happen, and you got through this with a Herculean effort, which is just so inspirational.

Surbhi: Totally. Yeah, the only thing I was ever without a doubt about truly was that patients needed this and that I wanted to be the one working on it.

Carolynn: Everything else, you probably second-guessed yourself all along the way.

Surbhi: All the time. All the time.

Jessica: Oh my God.

Carolynn: But at least you knew the mission.

Surbhi: Yes, and that was my guiding light.

Jessica: I love that. I know we're running out of time. I feel like I have questions that I wanted to ask you. I just have to say, I love... This is more of a comment and not a question, which I hate when people say that, but you said once, "I want to offer people who feel underrated a path to pursuing their dreams." That really spoke to me. I love that idea because in a way, that's what Y Combinator is.

Surbhi: And that's exactly what drew me to YC, that I could help all of these people who may not come from the most privileged backgrounds or may not have that faith in themselves yet, but I could help them see the best version of themselves and maybe be an example and maybe help them out on those early days, which is so difficult.

Carolynn: Yeah, for sure.

Jessica: Well, awesome. It has been so fascinating learning the details about nVision. The world needs to know more about this, and the world definitely needs to shine a bigger light on women's healthcare for sure.

Carolynn: Hear, hear.

Jessica: Congratulations. And it's been really, really fun learning about that.

Surbhi: Oh my gosh, this has been so much fun. Thank you so much for having me.

Carolynn: Thanks for coming, Surbhi.

Jessica: All right. We'll see you soon.

Surbhi: Thank you. See you soon.

Jessica: Bye.

Surbhi: Bye.

Carolynn: Her story, tons of struggle there, and I just think the stories where there's all this rejection, all this struggle, make the best stories, and she's got a great one.

Jessica: She just never gave up. That's incredible. And what a feat to be inventing a new device to insert into the fallopian tube, which is a millimeter wide.

Carolynn: I mean, everything about that story is how do you ever have success when you have that many, like we were saying, barriers before. There's all these barriers to success for that, and then you just keep on grinding away. That is incredibly impressive.

Jessica: And how she pointed out... And I think I knew this, but I didn't really realize it, about how women's health at that point just wasn't a thing when we are half the population, come on.

Carolynn: I think she said bikini health or something. It's crazy to me. But I mean, I think that's the story... That part was the least surprising that... Because women's health, completely outside of the entrepreneurial context, has always played second fiddle, so of course, combined with a female solo founder who, like she kept saying, didn't have the credentials that you might expect as selling it, she just had double whammy.

Jessica: My grandmother who raised me died of ovarian cancer.

Carolynn: Oh, really?

Jessica: And I think we've all probably known someone who's been touched by-

Carolynn: My grandma died of uterine cancer. So it's just very prevalent.

Jessica: I know. I want these devices that can just revolutionize the diagnosis and early.

Carolynn: I mean, I think we want it all to be a pap smear. If it can be that simple and that easy to do, if all the other parts could have that simple of a diagnostic tool, I think that's like Surbhi was explaining. That's what she wants.

Jessica: Yeah. Maybe she does have it in her someday for another woman's health startup.

Carolynn: We'll keep poking her.

Jessica: Yeah, yeah. Well, that was awesome. I loved it and I definitely would recommend her book, Without a Doubt. It's like a memoir/instructional-

Carolynn: Clearly she's been writing for a long time, so I would imagine the good writing combined with a good story makes for a good book.

Jessica: Yeah. Awesome. I think it's going to be a great episode when that comes out.

Carolynn: All right. See you next time. Bye.

Jessica: Okay, bye.